

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease:

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotypes

Student: Shahanta Dawn B. Balanza

Disease: Classic Galactosemia

Gene: GALT

Documented Variant: GALT c.563A>G (p.Gln188Arg/Q188R)

I. INTRODUCTION

1.1 Background

Genetic mutations are changes in DNA that can alter the information used to produce proteins. Depending on the type and location of a mutation, it may result in an amino-acid substitution, a premature stop codon, a change in the reading frame, or no change in the encoded protein. These molecular changes can affect protein structure and function and may contribute to genetic diseases. **Classic galactosemia** is an inherited metabolic disorder associated with reduced activity of the enzyme **galactose-1-phosphate uridylyltransferase (GALT)**, which is involved in the **Leloir pathway** for galactose metabolism. Pathogenic variants in the GALT gene can interfere with normal galactose processing and cause the accumulation of galactose and related metabolites.

In this laboratory investigation, the **GALT gene** and its association with classic galactosemia were examined at the DNA and protein sequence levels. A normal human GALT coding sequence was obtained from the **NCBI RefSeq database** and translated using **Galaxy** to establish a wild-type reference. A documented disease-associated variant, **GALT c.563A>G (p.Gln188Arg/Q188R)**, was then reproduced in the coding sequence and compared with the wild-type protein. A separate artificial single-nucleotide substitution was also introduced to examine how a different nucleotide change could affect the encoded protein.

The investigation focused on determining the relationship between a **DNA sequence change and its effect on the predicted protein sequence**. The wild-type and mutant sequences were compared to identify changes in amino-acid sequence, protein length, and other possible effects of the mutation. The results were then considered together with published scientific evidence to explain the potential molecular consequences of the **GALT c.563A>G**

(p.Gln188Arg/Q188R) variant and its association with classic galactosemia.

II. DISEASE AND GENE BACKGROUND

A. Disease

13. What is the disease or phenotype?

- Classic galactosemia is an inherited metabolic disorder caused by a deficiency of the enzyme galactose-1-phosphate uridylyltransferase (GALT). This deficiency prevents the normal breakdown of galactose, causing galactose-1-phosphate and other galactose-related compounds to accumulate in the body.

14. What are its major clinical characteristics?

- Major clinical characteristics include feeding difficulties, failure to thrive, jaundice, liver damage or liver failure, bleeding, and *Escherichia coli* sepsis in untreated infants. Long-term complications may include developmental and speech problems, motor abnormalities, cataracts, and premature ovarian insufficiency in females.

15. Which cells, tissues, or organs are mainly affected?

- The liver is an important organ affected because it is involved in galactose metabolism. Other tissues and organs can also be affected by the accumulation of galactose and its metabolites, including the nervous system, eyes, and reproductive system.

16. What is its genetic basis?

- Classic galactosemia is caused by pathogenic variants in both copies of the GALT gene. These variants reduce or eliminate GALT enzyme activity, impairing the metabolism of galactose.

17. What is its inheritance pattern, if applicable?

- Classic galactosemia follows an **autosomal recessive** inheritance pattern. An affected individual generally inherits a pathogenic GALT variant from each parent.

B. Gene and Normal Protein

18. What is the official gene symbol?

- The official gene symbol is **GALT**, which stands for **galactose-1-phosphate uridylyltransferase**.

19. On which human chromosome is the gene located?

- The **GALT gene is located on chromosome 9**, specifically at **9p13.3**.

20. What does the gene normally encode?

- The GALT gene encodes the enzyme **galactose-1-phosphate uridylyltransferase**, also

called GALT.

21. What is the normal biological function of the protein?

- The GALT enzyme helps the body metabolize galactose. It catalyzes a reaction in which **galactose-1-phosphate and UDP-glucose are converted into glucose-1-phosphate and UDP-galactose**. This allows galactose to be processed and used in normal cellular metabolism.

22. Where in the cell is the protein normally found?

- GALT is an **intracellular enzyme found in the cytosol**, where it carries out its role in galactose metabolism.

23. In what biological pathway or cellular process does it participate?

- GALT participates in the **Leloir pathway of galactose metabolism**, specifically catalyzing the second step of this pathway.

C. Documented Mutation

A well-documented pathogenic variant associated with classic galactosemia is **GALT c.563A>G (p.Gln188Arg)**, also known as **Q188R**. This variant is a single-nucleotide substitution in the **GALT** gene. It occurs in exon 6 and changes the nucleotide at coding position 563 from adenine (A) to guanine (G). This changes the codon from **CAG** to **CGG**, resulting in the replacement of glutamine (Q) with arginine (R) at amino-acid position 188. Therefore, the variant is classified as a **missense variant**. ClinVar classifies this variant as **pathogenic** for galactosemia. (NCBI ClinVar, 2026).

Table 1. Documented GALT c.563A>G (p.Gln188Arg) Variant Information

Variant Information	Details
Gene	GALT
Reference transcript	NM_000155.4
Reference protein	NP_000146.2
Exact variant notation	NM_000155.4:c.563A>G (p.Gln188Arg)
Nucleotide change	A → G
Codon change	CAG → CGG
Predicted protein change	p.Gln188Arg (Q188R)

Mutation type	Missense; single-nucleotide variant
Nucleotides affected	1 bp
ClinVar Variation ID	3614
ClinVar accession	RCV000825563.20
Clinical interpretation	Pathogenic
Associated condition	Galactosemia / Classic Galactosemia

The Q188R variant has been reported in individuals with galactosemia in both homozygous and compound-heterozygous states. Functional studies have also shown that the variant substantially reduces GALT enzyme activity. These findings support its classification as a pathogenic variant associated with classic galactosemia. (NCBI ClinVar, 2026; Elsevier & Fridovich-Keil, 1996).

III. NORMAL REFERENCE SEQUENCE

The normal reference coding sequence (CDS) for the **GALT** gene was obtained from the **NCBI RefSeq** database. The reviewed RefSeq transcript **NM_000155.4** represents GALT transcript variant 1 and encodes the reference GALT protein **NP_000146.2**. The CDS was selected for the manual translation experiment because it represents the protein-coding portion of the transcript.

Table 2. Reference Sequence Information for the GALT Gene

Reference Information	Details
Gene	GALT
Reference transcript	NM_000155.4
Reference protein	NP_000146.2
Sequence type used	Coding sequence (CDS)
Database	NCBI RefSeq
FASTA filename	GALT_WT_CDS.fasta

The normal GALT CDS was saved in FASTA format as **GALT_WT_CDS.fasta** and will be used as the wild-type (WT) sequence for the subsequent translation and mutation experiments.

IV. WILD-TYPE SEQUENCE ANALYSIS

The WT GALT CDS was uploaded into Galaxy and used as the reference sequence for the sequence analysis. A separate Galaxy history was created for the investigation to keep the analysis files organized. The WT sequence was retained without modification and was used as the control for the documented and artificial mutation experiments.

V. WILD-TYPE TRANSLATION

The normal **GALT** coding sequence (CDS) was used as the wild-type (WT) control. The WT CDS was translated into its predicted amino-acid sequence using the **SeqKit Translate** tool in Galaxy. The standard genetic code and reading frame 1 were used for translation. The resulting protein sequence was saved as **GALT_WT_protein.fasta**.

Table 3. Wild-Type GALT Sequence and Translation Results

WT Sequence Information	Result
Gene	GALT
Reference transcript/CDS	NM_000155.4
Reference protein	NP_000146.2
CDS length	1,140 bp
Predicted protein length	379 amino acids
Start codon	ATG
Stop codon	TAA
Reading frame	Frame 1
First 10 amino acids	MSRSGTDPQQ
Last 10 amino acids	QKDRETATIA
Protein FASTA filename	GALT_WT_protein.fasta

The translated WT CDS produced a predicted GALT protein sequence of 379 amino acids. The CDS is 1,140 bp long, corresponding to 380 codons, including the stop codon. The translation begins with the start codon ATG and ends with the stop codon TAA.

The predicted WT protein was compared with the accepted RefSeq GALT protein NP_000146.2. The predicted sequence agrees with the accepted reference protein in sequence

and length, confirming that the correct GALT CDS and reading frame were used. No sequence discrepancy was observed. Therefore, the WT sequence was established as the control for the subsequent documented mutation experiment.

VI. Mutation Hypothesis

Table 4. Predicted Effects of the GALT c.563A>G Mutation

Mutation Prediction	Expected Result
Mutation	GALT c.563A>G (p.Gln188Arg / Q188R)
Exact nucleotide change	A → G at coding position 563
Number of nucleotides affected	1 nucleotide (1 bp)
Predicted mutation type	Missense substitution / single-nucleotide variant
Predicted codon change	CAG → CGG
Predicted amino-acid change	Glutamine (Q) → Arginine (R) at position 188
Predicted effect on reading frame	No change; reading frame remains intact
Predicted effect on protein length	No change; expected to remain 379 amino acids
Predicted effect on protein function	Reduced or impaired GALT enzyme activity, potentially disrupting galactose metabolism

Hypothesis: If the documented **GALT c.563A>G** mutation is introduced into the WT CDS, then the resulting mutant protein is expected to remain the same length and reading frame but contain a **Q188R amino-acid substitution**, which is predicted to reduce GALT enzyme function and impair normal galactose metabolism.

VII. DOCUMENTED MUTATION EXPERIMENT

Table 5. Documented GALT c.563A>G Mutation Construction

Required Information	Result
Original nucleotide position	563
Original sequence	A

Mutant sequence	G
Original codon	CAG
Mutant codon	CGG
Bases inserted	0
Bases deleted	0
Bases substituted	1
Mutation type	Substitution (missense)

VIII. MUTANT SEQUENCE TRANSLATION

Table 6. Mutant GALT Sequence Translation Results

Required Information	Result
Mutant CDS length	1,140 bp
Mutant protein length	379 amino acids
Reading frame	Frame 1
First amino-acid difference	Amino acid 188 (Q → R)
Premature stop codon	Absent
Approximate number of amino acids affected	1 amino acid
Mutant protein FASTA filename	GALT_c.563A>G_mutant_protein.fasta

The mutant CDS remained 1,140 bp long and produced a predicted protein of 379 amino acids, the same length as the WT protein. The first sequence difference occurred at amino-acid position 188, where glutamine (Q) was replaced by arginine (R). No premature stop codon or reading-frame change was observed.

IX. WT AND MUTANT PROTEIN COMPARISON

24. At what amino-acid position do the sequences first differ?

- The WT and mutant protein sequences first differ at amino-acid position 188, where glutamine (Q) is changed to arginine (R).

25. Is only one amino acid affected?
- Yes. Only one amino acid is affected: **Q188R**.
26. Are multiple downstream amino acids changed?
- No. Multiple downstream amino acids are not changed; the sequences remain the same after position 188.
27. Was an amino acid deleted or inserted?
- No. No amino acid was deleted or inserted.
28. Was a premature stop codon produced?
- No. No premature stop codon was produced.
29. Did the reading frame change?
- No. The reading frame did not change because the mutation is a single-nucleotide substitution.
30. Did the protein length change?
- No. The protein length did not change. Both the WT and mutant proteins are 379 amino acids long.
31. Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?
- The mutation is a missense mutation, specifically GALT c.563A>G (p.Gln188Arg).

X. MOLECULAR CONSEQUENCE OF Q188R

The documented GALT c.563A>G mutation changes one nucleotide in the GALT coding sequence, replacing adenine (A) with guanine (G) at position 563. This changes the codon from CAG to CGG, resulting in a missense substitution of glutamine (Q) with arginine (R) at amino-acid position 188 (p.Gln188Arg/Q188R). The computational analysis showed that the mutation does not cause an insertion, deletion, or frameshift. The mutant protein remains 379 amino acids long, and no premature stop codon is produced.

Although only one amino acid is changed, the substitution can affect GALT protein structure and function. Published structural and biochemical studies indicate that Q188 is located near functionally important regions of GALT, including the active-site and dimer-interface regions. The replacement of glutamine with arginine can disrupt interactions involved in UDP-sugar binding and homodimer stability. Studies of Q188R have also reported increased aggregation and reduced ability of the enzyme to form the normal uridylylated intermediate, which is important for GALT catalytic activity (McCorvie & Timson, 2011; McCorvie et al., 2016).

Because GALT catalyzes the second step of the Leloir pathway, reduced GALT activity

interferes with the normal conversion of galactose-1-phosphate and UDP-glucose into UDP-galactose and glucose-1-phosphate. As a result, galactose-1-phosphate, galactose, and galactitol can accumulate, while reduced UDP-galactose availability can affect normal glycosylation processes (Berry, 2021).

Therefore, the molecular sequence of events can be summarized as:

GALT c.563A>G → CAG→CGG codon change → p.Gln188Arg (Q188R) → altered GALT structure/function → reduced GALT enzymatic activity → accumulation of galactose-1-phosphate and disruption of galactose metabolism → cellular and tissue effects associated with classic galactosemia.

The computational results demonstrate the DNA and protein sequence changes, while published biochemical and structural evidence supports the connection between the Q188R substitution and impaired GALT function. The exact severity of clinical manifestations can vary among individuals and is influenced by the specific GALT variant(s) and other factors (Berry, 2021).

XI. ARTIFICIAL MUTATION EXPERIMENT

A single-nucleotide substitution was introduced into the WT GALT CDS by changing c.564G>A, resulting in a codon change from CAG to CAA. Before translating the artificial mutant, the mutation was predicted to be synonymous because both codons encode glutamine (Q). Therefore, no change in the predicted protein sequence, protein length, or reading frame was expected.

Table 7. Artificial GALT c.564G>A Mutation Information

Artificial Mutation Information	Result
Gene	GALT
Reference transcript	NM_000155.4
Mutation type	Single-nucleotide substitution
Nucleotide position	c.564
Nucleotide change	G → A
Codon change	CAG → CAA
Predicted amino acid change	Q → Q

Predicted mutation type	Synonymous
Predicted frameshift	No
Predicted premature stop	No
Predicted protein length	379 amino acids

XII. COMPARISON OF DOCUMENTED AND ARTIFICIAL MUTATIONS

Table 8. Comparison of Wild-Type, Documented Mutant, and Artificial Mutant GALT Sequences

Feature	WT	Documented Mutation	Student-Created Mutation
CDS length	1,140 bp	1,140 bp	1,140 bp
Protein length	379 aa	379 aa	379 aa
Mutation	None	c.563A>G	c.564G>A
Codon change	CAG	CAG → CGG	CAG → CAA
Mutation type	N/A	Missense	Synonymous
Reading frame changed?	No	No	No
Premature stop codon?	No	No	No
Amino acids affected	None	Q188 → R188	None (Q → Q)
Expected functional consequence	Normal GALT protein function	May reduce GALT activity and affect protein structure/function	No change expected in the predicted protein sequence; functional effects were not tested.

The two mutations produced different outcomes because the nucleotide substitutions occurred at different positions within the codon. The documented **c.563A>G** substitution changed **CAG (Q) to CGG (R)**, producing a missense change at amino acid 188. In contrast, the artificial **c.564G>A** substitution changed **CAG (Q) to CAA (Q)**, and both codons encode

glutamine. Therefore, the artificial mutation did not alter the amino-acid sequence, while the documented mutation resulted in a change from glutamine to arginine. Both mutations retained the same CDS and protein lengths and did not produce a frameshift or premature stop codon.

XIII. Interpretation Questions

32. Why does the exact location of a mutation matter?

- The exact location matters because a mutation can affect different codons, amino acids, or functional regions of a protein. A change in an important region may have a greater effect on protein structure or function than a change in a less important region.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- Deleting three nucleotides can remove one amino acid while keeping the reading frame intact because three nucleotides make one codon. Deleting one or two nucleotides shifts the reading frame, changing how downstream codons are read.

34. Does every mutation change the amino-acid sequence? Explain.

- No. Some mutations are **synonymous**, meaning the changed codon still codes for the same amino acid. For example, in our artificial experiment, **CAG** → **CAA** still codes for glutamine (Q), so the protein sequence did not change.

35. Does every amino-acid substitution destroy protein function? Explain.

- No. The effect depends on the amino acid that is replaced, the new amino acid, and its location in the protein. Some substitutions have little or no effect, while others can alter protein structure, stability, or function.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- A single-nucleotide deletion shifts the reading frame. This changes how all subsequent nucleotides are grouped into codons, potentially changing many downstream amino acids and possibly creating a premature stop codon.

37. Why might a premature stop codon produce a nonfunctional protein?

- A premature stop codon ends translation before the complete protein is produced. The resulting shortened protein may lack important regions needed for proper folding, localization, or function.

38. Could a mutation affect protein function without greatly changing protein length?

- Yes. A single amino-acid substitution can affect an important active site, binding region,

or structural interaction while leaving the overall protein length unchanged. This is relevant to the documented **GALT Q188R** mutation.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes. A mutation can affect gene regulation, RNA splicing, mRNA stability, or the amount of protein produced without changing the amino-acid sequence. For example, a mutation in a regulatory region could reduce GALT expression and decrease the amount of functional GALT enzyme.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- Our analysis showed that **GALT c.563A>G** changes the DNA codon from **CAG to CGG**, producing the amino-acid substitution **Q188R**. The CDS remained **1,140 bp** and the protein remained **379 amino acids**, with no frameshift or premature stop codon. Published experimental studies are needed to support the further conclusion that this substitution reduces GALT enzyme function.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- Our computational results directly support the DNA and protein sequence changes, including the **c.563A>G** nucleotide substitution, **CAG→CGG** codon change, **Q188R** amino-acid substitution, unchanged protein length, and absence of a frameshift or premature stop codon. The conclusion that Q188R alters GALT structure, reduces enzyme activity, or contributes to the clinical features of classic galactosemia requires evidence from published biochemical, structural, and clinical studies.

XIV. GITHUB REPOSITORY AND GALAXY HISTORY

GITHUB REPOSITORY:

[balanzashahanta/Balanza_Galactosemia_GALT_Mutation_Lab: Investigation of the GALT mutation associated with classic galactosemia.](#)

GALAXY HISTORY:

<https://usegalaxy.org/u/shahantabalanza/h/balanza-galactosemia-galt-mutation-lab>

XV. REFERENCES

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